

DYNAMIC FRACTURE MECHANICS

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Abstract

Experimental data relating crack velocity to its purported driving force are reviewed and concepts put forth by Irwin are discussed. The computational convenience of such relation, which implies existence of a unique dynamic arrest stress intensity factor versus crack velocity relation as a material property, is noted. Crack propagation under tension and shear loadings is examined by the near field state of stresses for mixed mode loading. In the presence of large scale yielding in dynamic fracture, a quasi-static analysis using a modified Dugdale model was shown to adequately represent rapid tearing of thin ductile materials.

List of Symbols

A	Nondimensional constant ($\sqrt{r_c}(\sigma_{ox}/K_I)$)
a	crack length
δa	small segment of the x-axis at the crack tip
C	crack velocity
C_1	dilatational wave velocity
C_2	distortional wave velocity
E	Young's modulus
G	shear modulus
G	crack extension force
K	stress intensity factor
K_{Ia}	crack arrest toughness
K_{IC}	static plane strain fracture toughness
K_{Im}	minimum stress intensity factor
$K_{I_{dyn}}(t)$	time dependent stress intensity factor
$K_{I_{stat}}$	dynamic stress intensity factor
K_I	static stress intensity factor
K_{Ib}	branching stress intensity factor
K_{ID}	dynamic fracture toughness
K_I, K_{II}	mixed mode dynamic stress intensity factors
m	mixed mode stress intensity factor ratio, K_{II}/K_I
r, θ	radius and angle in polar coordinates
r_c	critical radial distance, material property
r_0	radial distance ahead of the crack tip tip at $\theta=0$
r_y	length of plastic yield zone
$\sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \sigma_{yz}$	rectangular cartesian stress components
$\sigma_{rr}, \sigma_{\theta\theta}, \sigma_{r\theta}$	stress components in polar components
σ_{ox}	remote stress or non-singular stress
σ_y	yield strength
τ_m	maximum shear stress
S	strain energy density factor
S_1, S_2	dimensionless functions of C/C_1 and C/C_2
ν	Poisson's ratio
ρ	density

Introduction

Although a start was made by Griffith over 60 years ago, fracture mechanics has largely evolved through the efforts of Irwin and his associates at the Naval Research Laboratory. Primitive crack arrest strategies were first employed in the 40's while work in the 60's and 70's tended to focus on controlling the initiation of crack growth for assuring structural integrity. However, there are conditions in which it would be too expensive to preclude crack instability under all possible service conditions and, further, where unstable crack propagation is completely unacceptable. The practical application of dynamic fracture mechanics is its role of arresting a rapidly propagating crack.

Research in dynamic fracture mechanics was not initiated until the mid 70's. Prior to such time, dynamic crack propagation was modeled with a sequence of static analysis and crack arrest was treated as the inverse of the onset of crack instability. Roberts and Wells¹ and Dulaney and Brace² used such static analysis together with Mott's dynamic extension³ of Griffith's instability criterion to represent the crack velocity in terms of the modulus of elasticity, the density and the critical crack growth. Berry^{4,5}, on the other hand, was the first to realize that a crack could propagate under a stress larger than the critical stress of Griffith and introduced the empirical multiplying factor which was greater than unity and which varied with the specimen boundary conditions. Since these energy theories were variations of the Griffith and Irwin's criterion, the resultant crack velocities were explicit functions of the initial crack growth. The early, but limited, experimental investigations⁶⁻⁸, which were conducted to establish design requirements for crack arresters in welded structures, unfortunately did not provide the necessary information which could be used

to assess the validity of the above-mentioned empirical relations on crack velocities.

Long before many solutions to elastostatic crack problems were available, Yoffe⁹ presented the steady state solution for a crack of constant length, moving along a straight line in an infinite two dimensional medium under remote tractions. The θ dependent portion of her solution for the crack-tip stress field, as shown in Figure 1, show that the maximum $\sigma_{\theta\theta}$ stress at the crack tip acted at an angle of 60 degrees to the crack propagation direction when the crack speed exceeded $0.62C_2$. Based on this finding, Yoffe suggested that at such high velocities, this change in the angle at which the maximum $\sigma_{\theta\theta}$ acts could cause the crack to deviate from its straight path.

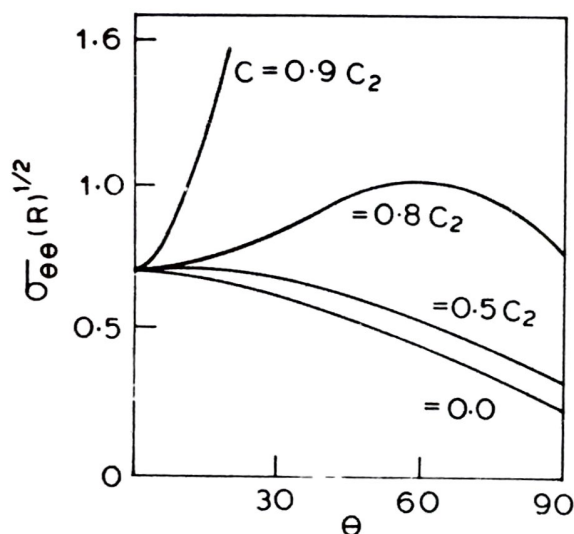


Fig.1. Angular variation of the circumferential stress [Yoffe]

Broberg¹⁰, Akita and Ikeda¹¹ and Mansinha¹², on the other hand, studied the problem of a central crack expanding at a constant rate in an infinite plane. Other elastodynamic solutions include a semi-infinite crack propagating at a constant velocity by Craggs¹³ and a propagating crack suddenly appearing in an infinite plane by Baker¹⁴. Validation of these theoretical solutions were stymied by the lack of detailed experimental data on the dynamic state of stress in the vicinity of the moving crack tip. In addition, available experimental data was influenced by the strong interactions between the reflected stress waves and the propagating crack tip in the fracturing specimens of finite geometries. Because of these theoretical and experimental difficulties, the pioneering dynamic photoelastic experiments by Wells and Post¹⁵, as well as other original experimental results of the late 60's¹⁶ and early 70's¹⁷ could not be used to verify the limited theoretical solutions.

Interest in dynamic fracture suddenly increased in the mid 70's with the concerns

for the safety of critical components in nuclear power plants subjected to seismic, projectile impact, and other accidental dynamic overloads. Also, an understanding of the physics of crack branching is required when assessing impact damages of fan blades, automotive and aircraft windshields; and in mining and oil recovery operations. There is considerable interest in the problem of arrest of a fast running crack in other large structures such as pipelines, ships, and airplanes. The timely developments in new analytical and experimental techniques coupled with the explosive advances in numerical techniques in fracture dynamics were used effectively to study the basic mechanics involved in rapid crack propagation. While controversies remain, plausible dynamic fracture criteria for a constant velocity crack, dynamic crack arrest, dynamic crack curving and dynamic crack branching have been proposed. A dynamic fracture criterion in the presence of large scale yielding is also available. The obvious application of these dynamic fracture criteria for predicting structural failure under impact loading is, however, hindered by the complexity of the dynamic state of stress in actual structural components.

The fundamental concepts of dynamic fracture and crack propagation discussed by Irwin is given in References 18-20; dynamic fracture of metals due to spalling is reviewed exhaustively by Meyers and Aime in Reference 21; and the developments in rapid crack propagation as discussed by Rose, in ref.22.

This paper will review these dynamic fracture criteria and will discuss their roles in analyzing fracturing structural components subjected to static or dynamic loadings.

Dynamic Fracture Criterion

Current work in dynamic fracture mechanics is primarily based upon the assumption of linear elastic material behavior, and can be generalized in the nomenclature of linear elastic fracture mechanics (LEFM). Two general concepts which have played a major role in the study of dynamic crack propagation are those of the dynamic stress intensity factor and the dynamic energy release rate. In this section, the mathematical definitions of these quantities are given in terms of the elastodynamic stress and deformation fields which prevail in a body of isotropic elastic material during crack propagation. The expressions for dynamic crack tip stresses are given in the next section. The parameter of most prominence is the time dependent stress intensity factor K , described by Irwin¹⁸ as:

$$K_I(t) = \lim_{r \rightarrow 0} (2\pi r)^{1/2} \sigma_{yy}(r, \theta, t) \text{ at } \theta = 0 \quad (1)$$

where σ_{yy} is the time dependent normal stress under the mode I loading condition. Definitions of K for mode II and III stress fields are obtained by replacing σ_{yy} , respectively, by shear stresses σ_{xy} and σ_{yz} in Equation (1). The definition of K is

meaningful only for linear elastic homogeneous materials which exhibit the $r^{-1/2}$ singularity at the tip of a sharp crack. Equation (1) holds for dynamic conditions because the nature of the singularity does not change under rapid crack propagation.

Similarly, the mode I crack extension force G , is defined as:

$$G = \lim_{a \rightarrow 0} \left\{ \frac{1}{\delta a} \int_0^{\delta a} v \sigma_{yy} dx \right\} \quad (2)$$

where v is the y -direction time dependent displacement of a point on the crack surface at a distance, δa behind the crack tip. Definitions of G for modes II and III stress fields can be obtained by replacing the integrand of Equation (2), by $u \sigma_{xy}$ and $w \sigma_{yz}$, respectively, where u and w are time dependent displacements in the x and z directions. In other words, G is the rate of energy loss at the crack tip singularity per unit of new separational area.

For a propagating crack, the fracture criterion is assumed to take the form in terms of K as:

$$K_I(t) = K_{ID}(C) \quad (3)$$

where C is the crack-tip speed. This hypothesis appears to be consistent with the fracture behavior of real materials. The actual dependence of K_{ID} on C must be determined by experiment.

Analytical difficulties associated with dynamic crack propagation have limited the solutions available to idealized situations. For mode I crack propagation at a nonuniform rate, it was shown²³ that the dynamic stress intensity factor depends on crack motion only through the instantaneous crack length and crack tip speed according to:

$$K_I^{\text{dyn}}(a, C) = k(C) K_I^{\text{stat}}(a, 0) \quad (4)$$

where k is a universal function of the crack speed C and K_I is the static stress intensity factor. This general result is valid only for the case of a semi-infinite crack in an infinite elastic medium. For a finite body, the result remains valid until reflected stress waves transmitting information about the changing boundary conditions reach the crack tip. This major limitation in the applicability of Equation (4) makes both numerical methods and direct experimental observations necessary for the complete understanding of the dynamic fracture process in finite bodies. The purpose of such experimental studies is to establish the existence of a $K_{ID}(C)$ versus C relationship and subsequently, to obtain its functional form for specific materials. Furthermore, once such a relationship is established, the value of the stress intensity factor at the instant of arrest is automatically determined.

Fracture Analysis

Analysis of dynamic crack propagation and crack arrest, for specific material, flaw geometry, and loading condition is broadly categorized below into Experimental Methods and Finite Element Method.

Experimental Methods

Over the past two decades, optical techniques have been used extensively to analyze the dynamic state of stress surrounding a rapidly propagating crack. Dynamic stress intensity factors were then computed by fitting a postulated static or dynamic crack tip field to the whole field experimental data which was generated by an optical technique. Only limited data on fracturing brittle metal²⁴⁻²⁷ was extracted using the two popular optical techniques, i.e., dynamic caustics and dynamic photoelastic coatings due to the experimental difficulties involved. Thus, much of the published experimental data^{15,24,25-41} related to the dynamic responses of photoelastic polymeric sheets.

Stress Intensity - Crack Velocity - Relationship

Figures 2 and 3 show the dynamic stress intensity factors, K_{ID} , versus crack velocity, C , relations of the two most studied photoelastic polymers; an epoxy resin of Araldite B⁴² and a polyester resin of Homalite-100³⁴. Figure 4 shows the K_{ID} versus C relations, which were obtained by different investigators, of Homalite-100. The K_{ID} versus C relation of Dally in Figure 4 was obtained by averaging the curves in Figure 3. While the K_{ID} versus C relations in these three figures exhibit some degree of specimen dependence, the crack will generally propagate at a constant velocity when driven by a dynamic stress intensity factor which exceeds the fracture toughness, K_{IC} . Variations in the terminal velocity, as shown in these figures, are due to the interaction between the reflected stress waves in the specimen and the propagating crack tip. Such stress wave effects, can be reduced, if not totally eliminated, in the one point bend specimen which is essentially a free-flying single edge notch beam³⁷.

Similar dynamic fracture toughness, K_{ID} , versus crack velocity, C , relations, which were obtained by direct measurements on fracturing metallic specimens, are few. Figure 5 shows such relation obtained by caustics for 4340 steel⁴⁴. Some data on K_{ID} versus C relation have been generated by a hybrid experimental-numerical procedure, described in a later section. Figures 6 and 7 show such data for A533 steel⁴⁵ and a ship hull steel KAS⁴⁶. Sufficient data, which may exhibit the pronounced specimen dependence noted in Figures 2 through 4, is lacking in these materials.

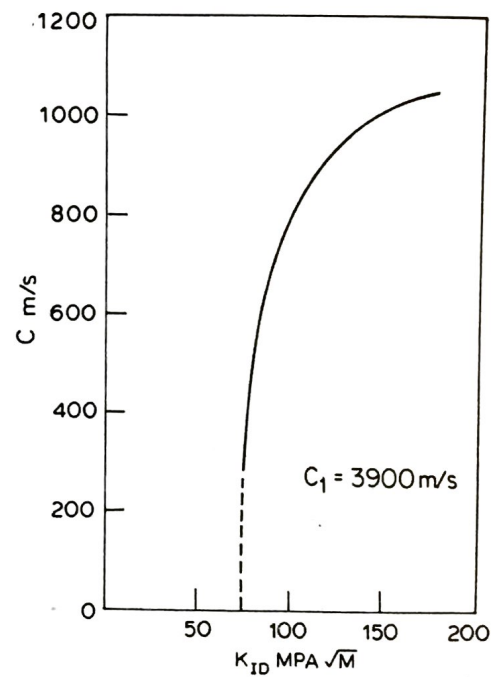
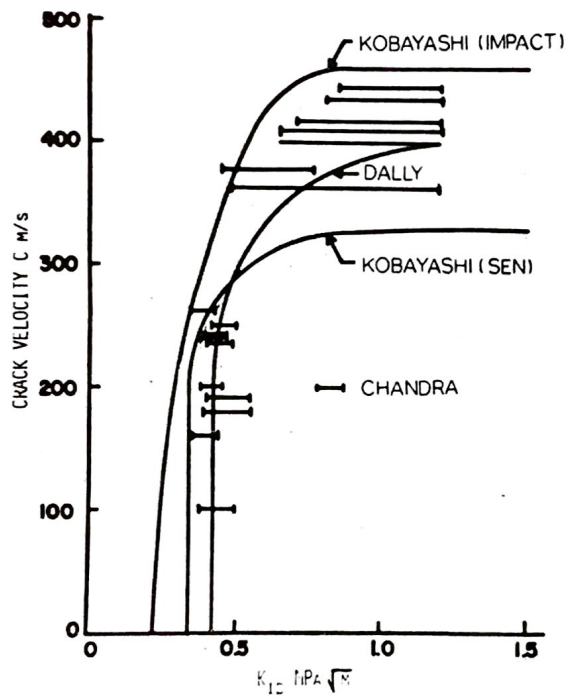
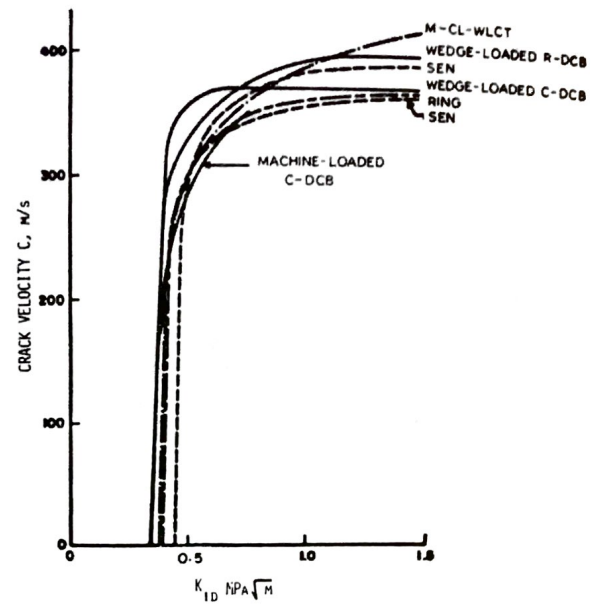
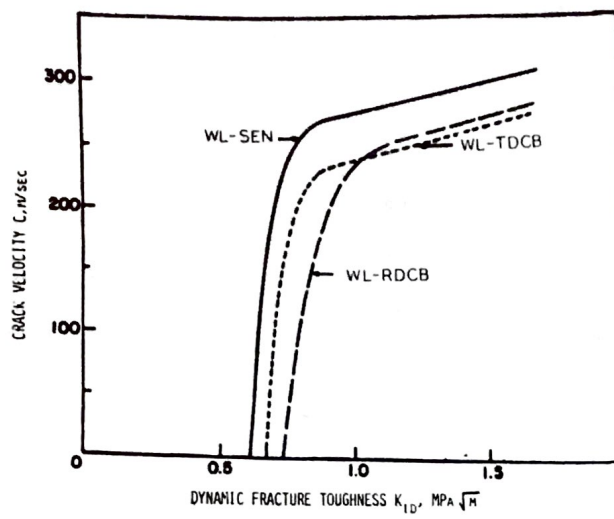
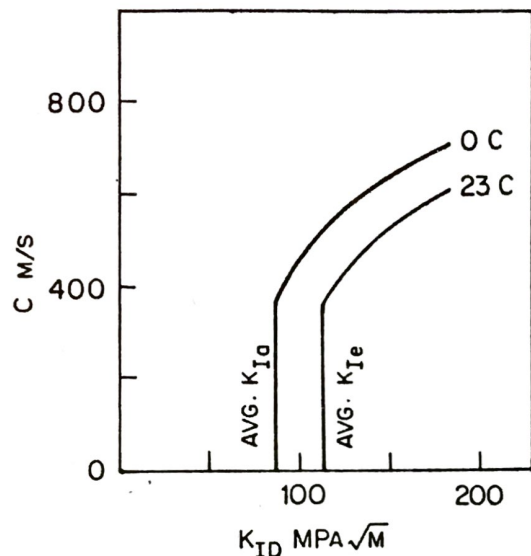
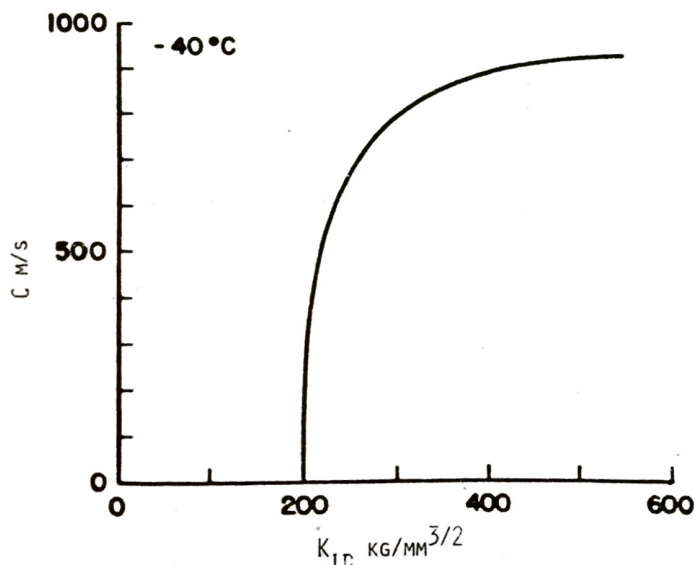


Fig.2. K_{ID} vs. C relations for WL-SEN, WL-TDCB and WL-RDCB specimens Araldite B. (Kalthoff et al).

Fig.3. K_{ID} vs. C relations Homalite-100 (Dally)

Fig.4. K_{ID} vs. C . relations, Homalite-100

Fig.5. K_{ID} vs. C 4340 steel (843° oil quench 315°C) (Rozakis et al)

Fig.6. K_{ID} vs. C A533B steel (Hahn et al.).Fig.7. K_{ID} vs. C KAS steel (Kanazawa et al.)

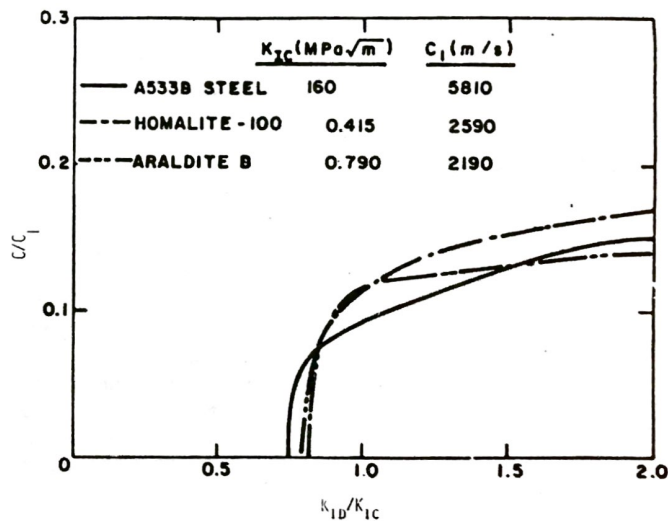
Crack Arrest

Characterization of crack arrest was made by static and kinetic arrest theories. The static arrest theory was rationally put forth by Irwin and Wells⁴⁷ in 1965 as:

It would now appear that the majority of running brittle cracks in mild steels are manifestations of plane strain fracture, such that arrests are simple reversals on the time scale of possible plane strain conditions.

This observation suggest that there exist a material property, K_{Ia} , that governs crack arrest in the same sense as K_{Ic} governs initiation. On the other hand, a kinetic approach focuses on crack propagation process and considers arrest only as the termination of such a process. Following these definitions, a notable features in Figures 2 through 7 is the apparent existence of a minimum dynamic stress intensity factor below which the crack ceases to propagate. This minimum stress intensity factor is often referred to as the dynamic arrest stress intensity factor, K_{Ia} , which at the most, is equal to the fracture toughness. Figures 2, 3 and 4 also show that to a lesser extent, K_{Ia} also is influenced by the stress wave interaction effect which can be predicted only by a systematic dynamic analysis.

When the average K_{ID} versus C relations of Figures 2 and 3 are normalized by their respective static fracture toughness, K_{Ic} and their dilatational wave velocities, C_1 , the resultant curves shown in Figure 8 become remarkably similar. Also shown in Figure 8 is the normalized K_{ID} versus C relation for A533B steel of Figure 6. The difference in the gamma-shaped (Γ), normalized K_{ID} versus C relations of polymers and steel are unexpectedly small.

Fig.8. Normalized K_{ID} vs. normalized C relations.

The above brief review of the K_{ID} versus C relations of polymers and steel indicate that within a generous engineering tolerance, these relations can be treated as a unique material property. In contrast to these generally accepted findings in steel and polymers, recent experimental evidence shows that the terminal crack velocity in structural ceramic is preceded by a crack acceleration phase. Also, the running crack will continue to propagate under a dynamic stress intensity factor which is much less than its fracture toughness^{48,49}. Figure 9 shows the normalized dynamic stress intensity factor K_{ID}^{dyn} versus C relation of a reaction bonded silicon nitride together with an idealized K_{ID} versus C relation of a typical turbine disk steel. Due to the absence of the characteristic gamma-shaped relation, which yields unique relations between

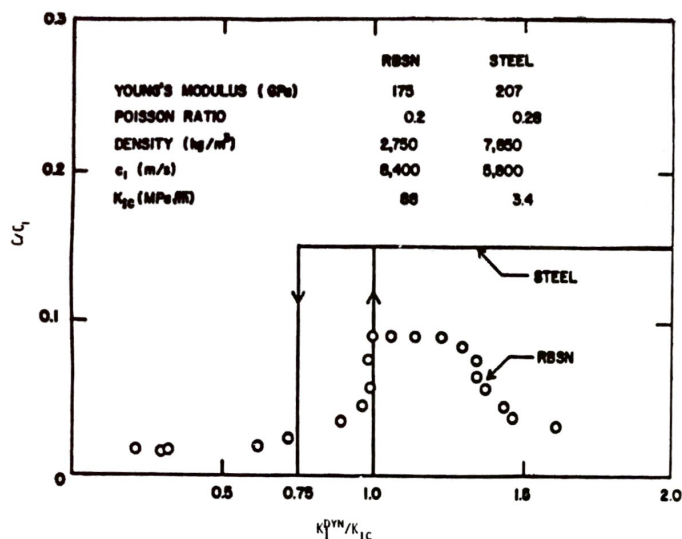


Fig.9. K_I^{dyn} vs. C reaction bonded silicon nitride and steel

K_{ID} and C in polymers and steel, K_I^{dyn} was used in place of K_{ID} for the structural ceramics in Figure 9. The lack of a unique K_I^{dyn} versus C relation as a material property also may indicate that the well established analytical techniques in dynamic fracture cannot be used to predict the dynamic fracture response of ceramic structures.

Finite Element Method

Lack of experimental procedures for directly measuring the precise dynamic parameters governing crack propagation and arrest in metals requires the use of an indirect procedure for determining dynamic fracture toughness, K_{ID} , and crack arrest stress intensity factor, K_{Ia} , in practical structural materials. The numerical procedure developed at Battelle Columbus Laboratories^{50,51} which is termed "Propagation Study", is based on comparative studies between experimental and numerical results using various assumed K_{ID} versus C relation. When the measured and calculated crack position versus time curves coincide, the assumed K_{ID} versus C relation is considered correct.

Another procedure, which is termed "Generation Study", uses the measured crack velocities to drive a crack and compute the resultant dynamic stress intensity factor based on the theory of dynamic elasticity. This procedure has been developed by the author and his colleagues^{52,53} and independently by Kanazawa and his colleagues⁵⁴.

Ignoring the above anomaly of structural ceramics for the time being, the experimentally observed maximum terminal crack velocity together with the observed crack arrest stress intensity factor can be incorporated into a finite element model to predict the dynamic fracture response of a steel (or polymeric) structure. In the following, examples of such dynamic fracture analyses are discussed.

For convenience in numerical analysis, the different normalized dynamic fracture properties of A533B steel and photoelastic polymers are modeled by the straight lines identified as steel in Figure 9. A single edge notch (SEN) specimen with a linearly varying displacement (constant end rotation) is considered. Figure 10 shows the finite element breakdown of the basic SEN specimen. The same specimen also was scaled to ten and a hundred times its original geometry in this dynamic fracture analysis. A dynamic finite element code was then executed in its "propagation phase"⁵⁰⁻⁵² by propagating the crack according to the idealized and normalized dynamic fracture toughness, K_{ID} , versus crack velocity, C , relation in Figure 9. Details of this dynamic finite element code and its use in solving dynamic fracture problems are given in References 52, 53 and, therefore, will not be repeated here. The procedure used in this propagation analysis is to first load the specimen until the static stress intensity factor, K_I , of the pre-existing edge crack reaches the prescribed fracture toughness, K_{IC} . The crack is then propagated one finite element node at a time following the prescribed nodal release mechanism⁵² and the dynamic stress intensity factor is computed from the energy dissipated at the released crack tip node.

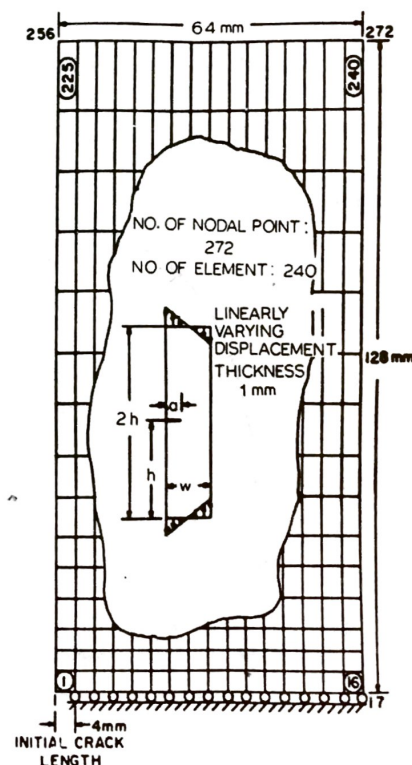


Fig.10. SEN specimen subjected to end rotation

Dynamic fracture analyses of A533B steel and Homalite-100 SEN specimens, of 64x256 mm as well as 640x2560 mm, ten times the size of the SEN specimens considered in Figure 10, also were analyzed. The resultant K_I^{dyn}/K_{IC} versus normalized

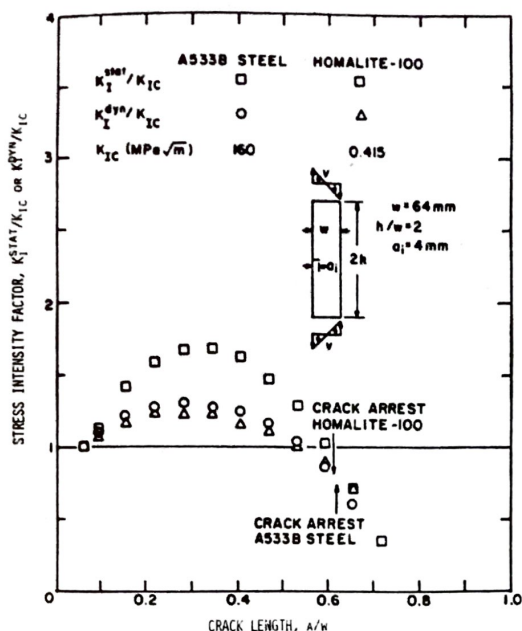


Fig.11. SEN specimen subjected to end rotation

crack length, a/W , relations of the regular as well as one hundred time SEN specimens can be superposed on the K_I^{dyn}/K_{IC} curve of Figure 11 with negligible difference. Agreements between the K_I^{dyn}/K_{IC} histories in A533B steel and Homalite-100 in Figure 11 are remarkable since Homalite-100, whose dilatational wave velocity is less than half of that of A533 steel, has a fracture toughness about 1/385 of that of A533B steel.

The above conclusion, which obviously is in need of further verification, indicates that model testing can be used to predict the dynamic fracture response of a prototype structural component provided the normalized K_{ID}/K_{IC} versus C/C_1 relations of the model and prototype materials coincide, and that scaled geometries are used. Thus the dynamic fracture response of an Araldite B model can be used to predict the corresponding fracture response of an A533B steel prototype component. Excluded from this scaling is the critical question involving crack arrest where the K_{ID}/K_{IC} values of the A533B steel and Araldite B materials, as shown in Figure 9, differ. For each investigation, a material which model more precisely the knee of the K_{ID}/K_{IC} versus C/C_1 relation of Araldite B as well as A533B steel, must be used.

The above conclusion is based on the extensive use of numerical analysis which in itself requires verification. While no formal benchmark problem in 2-D dynamic fracture was ever designated for round robin studies, an informal comparative study was made between the investigators at the Georgia Institute of Technology^{55,56}, Battelle Columbus Laboratories⁵¹, and the University of Washington⁵², each of which used different algorithms. The informal round robin analysis consisted of a dynamic crack propagation study of an A533B steel, dynamic tear test (DTT) specimen by generation calculation using the crack velocity data provided in ref.51. Figure 12

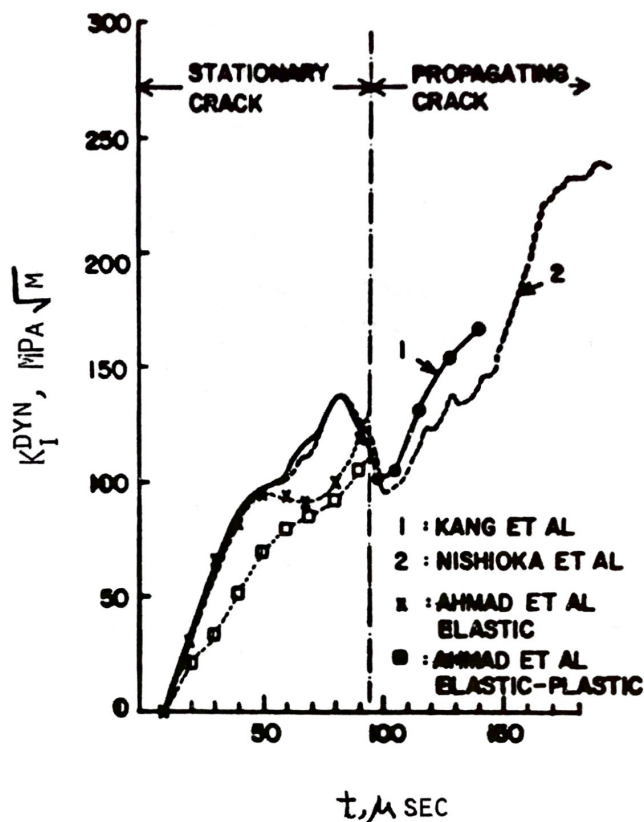


Fig.12. Dynamic stress intensity factor of a dynamic tear test specimen

shows the dynamic stress intensity factor, K_I^{dyn} , computed by the three investigators. All three elastic results are in good agreement with each other.

Returning to Figure 9, the differences in the dynamic fracture responses of steel and reaction bonded silicon nitride (RBSN) can be used to explain the observed differences⁵⁷ in failure modes of gas turbine discs fabricated from these two materials. The idealized disk geometry used for demonstration is shown in the caption of Figure 13. These disks are loaded with centrifugal forces generated by a constant spin velocity and with thermal stresses generated by a steady-state thermal gradient. In the following, the dynamic fracture response under a combined centrifugal and thermal stress loading is discussed.

For the dynamic fracture analysis of a steel disk, a single edge crack with an initial crack length of 5.1 mm, as shown in the caption of Figure 13, was assumed. The crack tip is at the location of the maximum stress intensity factor associated with the steady state thermal gradient and represents a crack which grow during the initial thermal shocking of the start-up stage of the gas turbine disk. The dynamic finite element analysis is simplified by the idealized K_I^{dyn} versus C relation where the crack propagates at a constant velocity of 15 percent of the dilatational stress wave velocity regardless of the driving force of K_I^{dyn} , as shown in Figure 9. This K_I^{dyn} independent crack velocity also

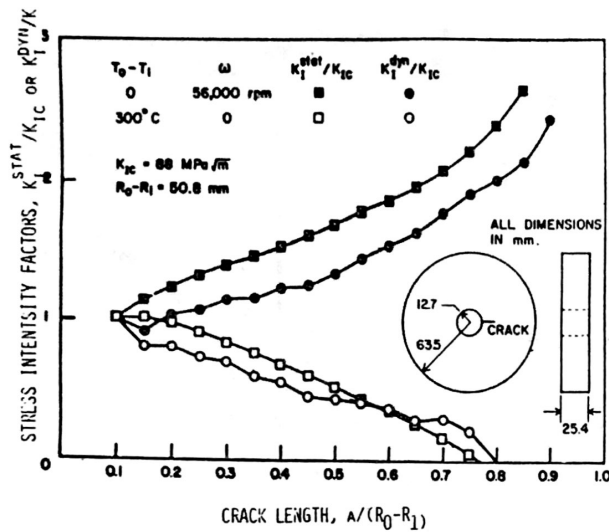


Fig.13. Normalized stress intensity factors for a steel gas turbine disk subjected to centrifugal force and thermal gradient

decouples the dynamic fracture analysis of the combined loading of centrifugal force and thermal gradient. The two dynamic stress intensity factors, which are obtained separately through two dynamic fracture analyses of the disk subjected to centrifugal force and to thermal gradient, are then superposed to yield the resultant dynamic stress intensity factor for the combined loading.

Figure 13 shows the normalized dynamic stress intensity factors for the steel disk subjected to centrifugal force and to thermal gradient loadings, respectively. These results were obtained by the generation mode of dynamic finite element method where the crack was driven at the terminal velocity of $0.15 C_1$ regardless of the magnitude of K_I^{dyn} . Also shown for comparison are the corresponding static stress intensity factors for the two loadings. These results then can be superposed to construct the resultant dynamic stress intensity factors of the steel disk subjected to various combinations of spin velocity and thermal gradients.

The relatively slow crack velocity of reaction bonded silicon nitride (RBSN), as shown in Figure 9, allows the use of quasi-static analysis to study the crack arrest potential of a RBSN disk. Since a rapidly propagating crack in reaction bonded silicon nitride will only arrest under a vanishing static stress intensity factor, the results of Figure 13 show that the crack will not arrest under most of the spin velocities.

Near Field State of Stresses of a Rapidly Running Crack

The general crack tip dynamic state of stress under mixed mode conditions was given recently by Nishioka and Atluri⁵⁸ in terms of the local rectangular (x,y) and polar (r,θ) coordinates. The three

rectangular stress components under mode I and mode II conditions are given as

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix} = \sum_{n=1}^{\infty} A_{In} \frac{B_I(c)}{\sqrt{2\pi}} \frac{n(n+1)}{2} \times$$

$$\begin{bmatrix} (1+2S_1^2-S_2^2)r_1^{(n/2-1)} \cos(\frac{n}{2}-1)\theta_1 - 2h(n)r_2^{(n/2-1)} \cos(\frac{n}{2}-1)\theta_2 \\ -(1+S_2^2)r_1^{(n/2-1)} \cos(\frac{n}{2}-1)\theta_1 + 2h(n)r_2^{(n/2-1)} \cos(\frac{n}{2}-1)\theta_2 \\ -2S_1r_1^{(n/2-1)} \sin(\frac{n}{2}-1)\theta_1 + \frac{(1+S_2^2)}{S_2} h(n)r_2^{(n/2-1)} \sin(\frac{n}{2}-1)\theta_2 \end{bmatrix}$$

$$+ \sum_{n=1}^{\infty} A_{II n} \frac{B_{II}(c)}{\sqrt{2\pi}} \frac{n(n+1)}{2} \times$$

$$\begin{bmatrix} (1+2S_1^2-S_2^2)r_1^{(n/2-1)} \sin(\frac{n}{2}-1)\theta_1 - 2h(\bar{n})r_2^{(n/2-1)} \sin(\frac{n}{2}-1)\theta_2 \\ -(1+S_2^2)r_1^{(n/2-1)} \sin(\frac{n}{2}-1)\theta_1 + 2h(\bar{n})r_2^{(n/2-1)} \sin(\frac{n}{2}-1)\theta_2 \\ 2S_1r_1^{(n/2-1)} \cos(\frac{n}{2}-1)\theta_1 - \frac{(1+S_2^2)}{S_2} h(\bar{n})r_2^{(n/2-1)} \cos(\frac{n}{2}-1)\theta_2 \end{bmatrix} \quad (5)$$

where

$$S_1^2 = 1 - \frac{C^2}{C_1^2}; \quad S_2^2 = 1 - \frac{C^2}{C_2^2}$$

$$B_I(C) = \frac{1+S_2^2}{D(C)}; \quad B_{II}(c) = \frac{2S_2}{D(C)}$$

$$D(C) = 4S_1S_2 - (1+S_2^2)^2$$

$$h(n) = \begin{cases} 2S_1S_2/(1+S_2^2) & : n \text{ odd} \\ (1+S_2^2)/2 & : n \text{ even} \end{cases}$$

$$h(\bar{n}) = h(n+1)$$

$$r_1^2 = x^2 + S_1^2 y^2$$

$$r_2^2 = x^2 + S_2^2 y^2$$

$$r^2 = x^2 + y^2$$

$$\tan \theta_1 = S_1 \tan \theta$$

$$\tan \theta_2 = S_2 \tan \theta$$

$$\tan \theta = y/x$$

r and θ are the polar coordinates with origin at the moving crack tip and C, C₁, and C₂ are the crack velocity, dilatational, and distortional wave velocities, respectively.

The general solution expressed in Equation (5) yields the singular stresses when $n=1$; i.e., $A_{II1}=K_{II}^*$ and $A_{II2}=K_{II}^*$, which are stress intensity factors of mode I and mode II, respectively. The constant stress, σ_{ox} , is related to the higher order terms (HOT) for $n \geq 2$ as:

$$\sigma_{ox} = \frac{6B_I(C)(S_1^2 - S_2^2)}{\sqrt{2\pi}} A_{I2} \quad (6)$$

$$A_{II2} = 0$$

The $n=2$ term exists only in σ_{xx} , stress component of mode I loading, and is zero in all other components of mode I and mode II stress fields. Under pure mode I loading conditions (i.e., $K_{II}=0$) and $n=1$, the state of stresses ahead of a rapidly running crack yields

$$\frac{\sigma_{xx}}{\sigma_{yy}} = \frac{(1+2S_1^2-S_2^2)(1-S_2^2)-4S_1S_2}{4S_1S_2-(1+S_2^2)^2} \quad (7)$$

and shows equality at zero crack velocity to increase at the Rayleigh speed.

Mixed-Mode Analysis of Moving Crack

The dynamic singular crack tip stress field for small θ values differ with the corresponding static stress field in that the largest principal singular tensile stress acts parallel to the x-axis not only in mode I loading but also in mixed mode loading for smaller values of K_{II}/K_I . Furthermore, the ratio of σ_{xx}/σ_{yy} is roughly equal to unity for both static and dynamic loads up to $C/C_1=0.10$. Situations arise however, for which a single parameter characterization, K_I or K_{II} , in mode I or mode II, crack tip stress field is not adequate due to spreading of the fracture process zone or reduction in size of the K-dominated singular zone. For example, even in a brittle solid, roughening of the fracture surface due to spreading of advance cracking and incipient branching can substantially enlarge the fracture process zone. As a second example, when dynamic photoelasticity is used to evaluate K_I and K_{II} , the isochromatic fringes very close to the crack tip normally cannot be used due to light scattering from the caustic at the crack tip, crack front curvature, and lack of fringe clarity. The experimental stress analyst is thus frequently forced to take measurements from regions which border the valid region of the above crack tip field equations. In such cases, it is desirable to incorporate an additional stress field parameter, σ_{ox} . Irwin, in 1958⁵⁹, introduced this remote stress, σ_{ox} , in order to match the experimental stress fields of reference 15. This inevitable involvement of the second order term forms the basis of incorporating σ_{ox} in dynamic mixed mode analysis of moving crack.

* The superscript "dyn" to identify dynamic stress intensity factor will not be used, since all quantities refer to dynamic values.

Dynamic Crack Curving

Crack extension and fracture criteria under combined tension and shear loading are based on either maximum circumferential stress or energy criteria. However, some investigators used strain energy density to predict the crack extension angle under combined loading^{60,63,65} and it is therefore reviewed in this section.

Maximum Circumferential Stress Criterion

The angle θ_c , at which circumferential stress, $\sigma_{\theta\theta}$, is maximum, can be obtained from the following:

$$\begin{aligned} \sigma_{\theta\theta} = & \frac{K_I B_I}{\sqrt{2\pi}} \left[((S_1^2 - S_2^2) - (1 + S_1^2) \cos 2\theta) \frac{\cos \frac{\theta_1}{2}}{\sqrt{r_1}} \right. \\ & + \frac{4S_1S_2}{1+S_2^2} \cos 2\theta \frac{\cos \frac{\theta_2}{2}}{\sqrt{r_2}} - \\ & \left. - 2S_1 \sin 2\theta \left(\frac{\sin \frac{\theta_1}{2}}{\sqrt{r_1}} - \frac{\sin \frac{\theta_2}{2}}{\sqrt{r_2}} \right) \right] \\ & + \frac{K_{II} B_{II}}{\sqrt{2\pi}} \left[-2S_1 \sin 2\theta \frac{\cos \frac{\theta_1}{2}}{\sqrt{r_1}} \right. \\ & + \frac{(1+S_2^2)^2}{2S_2} \sin 2\theta \frac{\cos \frac{\theta_2}{2}}{\sqrt{r_2}} + ((S_2^2 - S_1^2) + (1 + S_1^2) \cos 2\theta) \\ & \left. \frac{\sin \frac{\theta_1}{2}}{\sqrt{r_1}} - (1 + S_2^2) \cos 2\theta \frac{\sin \frac{\theta_2}{2}}{\sqrt{r_2}} \right] + \frac{\sigma_{ox}}{2} (1 - \cos 2\theta) \end{aligned} \quad (8)$$

$$\partial \sigma_{\theta\theta} / \partial \theta = 0 ; \quad \sigma_{\theta\theta} > 0$$

and added $\sigma_{\theta\theta} > 0$ is to assure fracture in tension under mixed mode loading. For a pure mode I dynamic crack tip state of stresses Equation (8) yield a transcendental relation between the critical values of θ and r ^{60,61}:

$$r = \frac{1}{4\pi} \left[\left(\frac{K_I}{\sigma_{ox}} \right) V(\theta, C, C_1, C_2) \right]^2 \quad (9)$$

Furthermore, by letting $\theta=0$ in Equations (9), we obtain:

$$r_0 = \frac{1}{128\pi} \left[\left(\frac{K_I}{\sigma_{ox}} \right) V_0(C, C_1, C_2) \right]^2 \quad (10)$$

and

$$\begin{aligned} V_0(C, C_1, C_2) = & [B_I(C) \{ -(1 + S_2^2)(2 - 3S_1^2) - \frac{4S_1S_2}{1+S_2^2} (14 + 3S_2^2) \\ & + 16(1 + S_2S_2) \}] \end{aligned}$$

at the zero crack velocity, $C/C_1=0$, Equation (10) reduces to Streit and Finnie's solution^{6,1} of:

$$r_0 = \frac{9}{128\pi} \left(\frac{K_I}{\sigma_{ox}} \right)^2$$

Directional instability is assumed to occur when the running deviates from its straight path at $r_0 \leq r_c$. Furthermore, r_c is assumed to be a material constant which can be determined experimentally by using Equation (10).

Minimum Strain Energy Density Criterion

The strain energy density criterion, dW/dV , developed by Sih^{6,3} is based on the local density of the strain energy field surrounding a crack. It assumes that fracture initiates when dW/dV reaches a critical value of $(dW/dV)_c$.

The elastic strain energy density, dW , per unit volume, dV , of the material is:

$$\frac{dW}{dV} = \frac{1+\nu}{2E} (\sigma_{xx}^2 + \sigma_{yy}^2 + \sigma_{zz}^2) - \frac{\nu}{1+\nu} (\sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx}) + 2\sigma_{xy}^2 \quad (11)$$

for plane strain condition:

$$\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy}) \quad (12)$$

and thus

$$\frac{dW}{dV} = \frac{1+\nu}{2E} [(1-\nu)(\sigma_{xx}^2 + \sigma_{yy}^2) - 2\nu\sigma_{xx}\sigma_{yy}] + 2\sigma_{xy}^2 \quad (13)$$

where E and ν are the modulus of elasticity and Poisson's ratio, respectively.

The energy density field dW/dV in the vicinity of the crack border may be expressed in the well known form:

$$dW/dV = \frac{S}{r} + \text{non-singular terms} \quad (14)$$

The coefficient of $1/r$ in the above is referred to as the strain energy density factor, S . Since the strain energy density, dW/dV , approaches infinity as r tends to zero, a finite dW/dV is evaluated at a critical distance r_c . The region inside r_c , where the material is expected to be nonlinear elastic and plastic, is called the core region.

The intensity of the strain energy density for the state of plane strain at a critical distance r_c can be written as:

$$S' = r_c(dW/dV)^{**} \quad (15)$$

^{**}Sih^{6,2} defined S with only the singular terms. S' , which incorporates both the singular and non-singular stresses.

$$S' = r_c \frac{(1+\nu)}{2E} [(1-\nu)(\sigma_{xx}^2 + \sigma_{yy}^2) - 2\nu\sigma_{xx}\sigma_{yy}] + 2\sigma_{xy}^2 \quad (16)$$

Substituting Equation (5) into Equation (12) for first two terms (i.e., $n=2$) yields:

$$S' = [a_{11}K_I^2 + 2a_{12}K_I K_{II} + a_{22}K_{II}^2] + \sqrt{r_c} \sigma_{ox} (a_{31}K_I + a_{32}K_{II}) + a_{33}(\sqrt{r_c} \sigma_{ox})^2$$

where

$$\begin{aligned} a_{11} &= \frac{1}{16\pi G} [(1-\nu)(F_{11}^2 + F_{22}^2) - 2\nu F_{11}F_{22} + 2F_{12}^2] \\ a_{12} &= \frac{1}{16\pi G} [(1-\nu)(F_{11}G_{11} + F_{22}G_{22}) - \nu(F_{11}G_{22} + F_{22}G_{11}) + 2F_{12}G_{12}] \\ a_{22} &= \frac{1}{16\pi G} [(1-\nu)(G_{11}^2 + G_{22}^2) - 2\nu G_{11}G_{22} + 2G_{12}^2] \\ a_{31} &= \frac{1}{4\sqrt{\pi G}} [(1-\nu)F_{11} - \nu F_{22}] \\ a_{32} &= \frac{1}{4\sqrt{\pi G}} [(1-\nu)G_{11} - \nu G_{22}] \\ a_{33} &= \frac{1}{4G} [1-\nu] \end{aligned} \quad (17)$$

$$F_{11}(C, C_1, C_2, \theta) = B_I(C) \left[(1+2S_1^2 - S_2^2)f_{11} - \frac{4S_1S_2}{(1+S_2^2)} f_{22} \right]$$

$$G_{11}(C, C_1, C_2, \theta) = B_{II}(C) \left[-(1+2S_1^2 - S_2^2)g_{11} + (1+S_2^2)g_{22} \right]$$

$$F_{22}(C, C_1, C_2, \theta) = B_I(C) \left[-(1+S_2^2)f_{11} + \frac{4S_1S_2}{(1+S_2^2)} f_{22} \right]$$

$$G_{22}(C, C_1, C_2, \theta) = B_{II}(C) \left[(1+S_2^2)g_{11} - (1+S_2^2)g_{22} \right] \quad (18)$$

$$F_{12}(C, C_1, C_2, \theta) = B_I(C) [2S_1(g_{11} - g_{22})]$$

$$G_{12}(C, C_1, C_2, \theta) = B_{II}(C) \left[\frac{1}{2S_2} \{ (4S_1S_2)f_{11} - (1+S_2^2)f_{22} \} \right]$$

$$f_{11} = [f(C_1) + g(C_1)]^{1/2}$$

$$g_{11} = [f(C_1) - g(C_1)]^{1/2}$$

$$f_{22} = [f(C_2) + g(C_2)]^{1/2}$$

$$g_{22} = [f(C_2) - g(C_2)]^{1/2}$$

$$f(C_1) = \frac{1}{(1 - \frac{C_2^2}{C_1^2} \sin^2 \theta)^{1/2}}; \quad g(C_1) = \frac{\cos \theta}{(1 - \frac{C_2^2}{C_1^2} \sin^2 \theta)}$$

$$f(C_2) = \frac{1}{(1 - \frac{C_2^2}{C_2^2} \sin^2 \theta)^{1/2}}; \quad g(C_2) = \frac{\cos \theta}{(1 - \frac{C_2^2}{C_2^2} \sin^2 \theta)}$$

By setting $\partial S'/\partial \theta$ equal to zero, the following expression is obtained:

$$m^2 \frac{\partial a_{22}}{\partial \theta} + m \left(\frac{\partial a_{12}}{\partial \theta} + A \frac{\partial a_{32}}{\partial \theta} \right) + \left(\frac{\partial a_{11}}{\partial \theta} + A \frac{\partial a_{31}}{\partial \theta} \right) = 0 \quad (19)$$

where

$$m = K_{II}/K_I$$

$$A = \sqrt{r_0} \sigma_{ox}/K_I$$

The fracture angle, θ_c , at which S' possesses stationary values can be obtained from Equation (19). Comparison between the static and the corresponding dynamic circumferential stress and elastic strain energy densities (ESED) for identical K_I and K_{II} is informative.

Equations (8) and (19) show that curving of a straight crack propagating at the lower velocity can be considered only by incorporating the non-singular term of σ_{ox} in the minimum strain-energy-density criteria.

Comparison of Maximum $\sigma_{\theta\theta}$ and Minimum S' Criterion

Figure 14 shows the predicted crack-curving angles for crack velocities, $0 \leq C/C_1 \leq 0.25$ by the maximum circumferential stress and the minimum strain-energy-density criterion with $\sigma_{ox}=0$. Without the second-order term, both criteria predicted the same crack-curving angles for much of the crack-velocity range. Figure 14 also shows that for lower crack velocities of $C/C_1 \leq 0.15$, the predicted crack-curving angle, hereafter referred to as fracture angle, is almost equal to the corresponding static fracture angles.

The effects of the non-singular term of σ_{ox} and reference distance r_0 , on the fracture angle predicted by both maximum $\sigma_{\theta\theta}$ and minimum S' criteria at various crack velocities showed that fracture angle for negative σ_{ox} is much smaller than those with positive σ_{ox} . Figure 15 shows the fracture angles predicted by S' criteria for a positive K_{II}/K_I at $C/C_1=0.2$ and Poisson's ratio of 0.33 in the presence of non-singular stress. Also, for a given tensile or compressive σ_{ox} , larger A results in larger fracture angle. Detailed discussion on this subject is given in References 60 and 64.

Dynamic Crack Branching

Propagating cracks in dynamically loaded structures often curve when impacted by a stress wave and branch under high crack tip loading. Clark and Irwin⁶⁶, in agreement with the experimental observations reported by Schardin⁸, found that the crack speed just before crack division was nearly the same as that of the most advanced crack in a multiple division crack pattern. In contrast to Yoffe's argument⁹, the crack does not branch solely because it has reached some critical velocity,

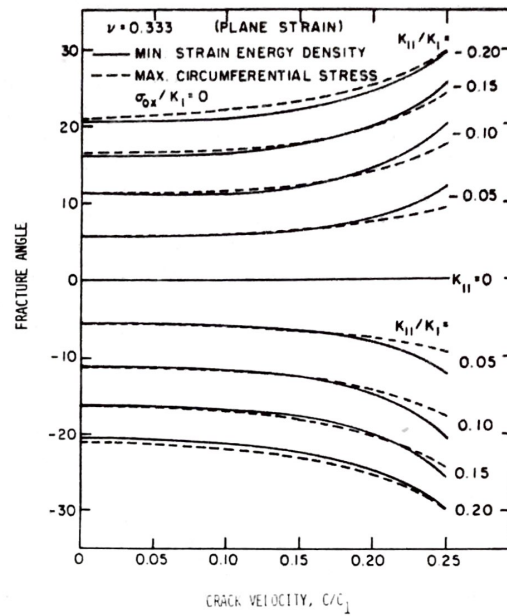


Fig.14. Fracture angle predicted by maximum circumferential-stress criterion and minimum strain-energy-density criterion, $\sigma_{ox}=0$.

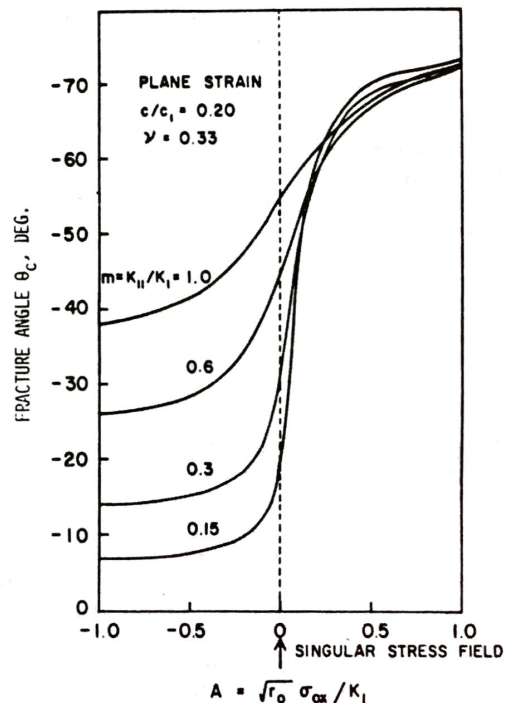


Fig.15. Effect of 'A' on predicted fracture angles. Mixed load.

as depicted in Figure 1. Clark and Irwin⁶⁴ were able to arrive at an alternative explanation through quasi-static estimates of the crack driving forces prior to and following branching. Upon finding them to be nearly equal, they concluded that

It seems best therefore to regard attainment of a critical K (or G), rather than a crack-speed-

induced modification of the stress pattern, as the primary factor controlling crack division.

This same conclusion was reached independently by Congleton and Petch⁶⁷ who offered a rationalization based on the nucleation of cracks ahead of the moving crack tip. In contrast to the observations reported by Clark and Irwin, however, their findings indicated that branching can occur at crack speeds well below the limiting value. Hence, another alternative proposed by the authors^{68,69} is a micro-mechanical model which couples the specimen behaviour in terms of K_{ID} , C , and non-singular stress, σ_{0x} , with the nucleation of secondary cracks ahead of the propagating crack tip. A dynamic crack curving criteria, which utilizes in addition to the mixed-mode dynamic stress intensity factors, the second-order term in the crack tip state of stress. The proposed model is verified by the authors through dynamic photoelastic analyses of propagating curved cracks in photoelastic polymers^{60,61}. It is a dynamic extension of the static crack curving criterion by Streit and Finnie⁶², and is based on the postulate that the micro-cracks ahead of the crack tip dictate the direction of crack propagation. An off-axis micro-void, which is within a critical distance, r_c , to the crack tip, is actuated by a critical crack tip stress field and deflects the crack from its otherwise self-similar propagation path. The distance between the micro-void and the crack tip, r_0 , is a characteristic distance which is governed by the singular state of stress as well as the second order term, i.e., the remote stress component. The critical distance, r_c , is assumed to be a material property.

The associated crack branching criterion^{68,69}, requires a critical stress intensity factor to trigger crack branching and a crack curving criterion to predict the crack branching angle. A review of all past and recent crack branching data on Homalite-100 and polycarbonate photoelastic polymers showed that most crack would bifurcate upon reaching a branching stress intensity factor. The bifurcated cracks also satisfied the crack curving criterion which is the sufficiency condition for the bifurcated cracks to propagate in a slanted direction.

Figure 16 shows the variations in dynamic stress intensity factors, K_I and K_{II} , and the characteristic distance, r_0 during crack branching in a Homalite-100, single-edged notch specimen. Table 1 shows the crack branching stress intensity factors, K_{Ib} , the critical distance, r_0 and the measured and the predicted crack branching angles. Crack branching angles were computed by using the crack curving criterion. Similar results have been obtained from a recently completed dynamic photoelastic analysis of crack branching in polycarbonate single-edged notch specimens⁷⁰. The crack branching criterion was also used to predict the crack branching angle of bifurcated cracks which initiated under static loading at the blunt starter cracks in Homalite-100, wedge-loaded, double-cantilever beam specimens. The crack branching angles in pressurized steel^{71,72} and aluminium⁷³ pipes, as shown in Table 2, were also predicted by this crack branching criterion. While these criteria are easily incorporated into

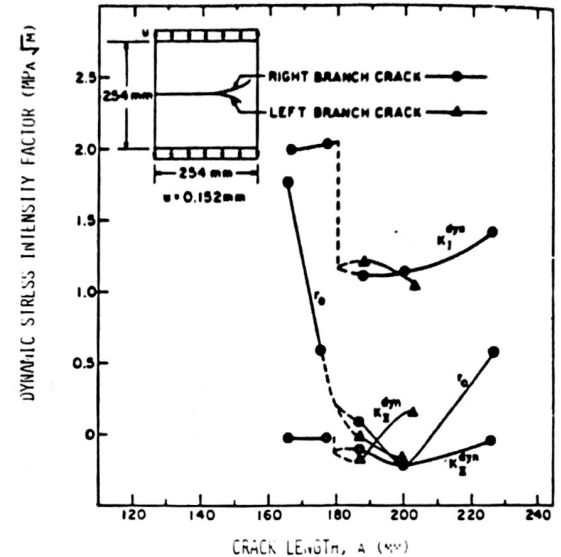


Fig.16. Dynamic stress intensity factors and r_0 of branched cracks Homalite-100 SEN specimen.

Table 1
Crack branching data
Homalite-100 single-edged notch specimens

Specimen No.	Branching K_{Ib} MPa√m	r_c mm	Branching angle	
			Measured	Estimated
B8	2.08	1.2	23	28
B9	2.03	1.3	30	28
W08270	2.03	1.4	26	28
B7	2.03	1.4	30	28
B5	2.08	1.2	30	28
B6	2.09	1.3	28	28
Average	2.04	1.3	27.8	28

Table 2
Crack branching data
(pressurized metal pipes)

Material	Branching K_{Ib} MPa√m	r_0 mm	Branching angle	
			Measured	Predicted
Steel	124	1	66	64
Aluminium	xxx	1.3	88	84

a finite element model, the actual algorithms which model crack curving and nonsymmetric crack branching require special procedures, such as double nodding and node grafting.

Rapid Tearing of Polycarbonate Sheets

Recent dynamic photoelastic analysis of fracturing 1.6 mm thick polycarbonate sheets has shown

that the crack velocity is less than 5 percent of the dilatational stress wave velocity and that the propagating crack tip is preceded by a necked region. The lower crack velocity reduced dynamic fracture of thin polycarbonate sheets to quasi-static analysis and the latter justified the use of a Dugdale strip yield zone. A Dugdale strip yield zone together with two tangential surface tractions, which modeled the residual stresses in the wake of rapid tearing, generated isochromatics which resembled the isochromatics surrounding the propagating crack⁷⁴.

Moving Dugdale Strip Yield Zone

The static and dynamic near-field states of stress in the vicinity of a Dugdale strip yield zone, as shown in Figure 17, were derived in ref.75 and 27 & 76 respectively. In terms of the local polar coordinate system shown in Figure 17, the dynamic maximum shear stress, τ_m , can be represented in the three stress components σ_{xx} , σ_{yy} , and σ_{xy} , which were derived from ref.77 as

$$\tau_m = \left[\left(\frac{\sigma_{yy} - \sigma_{xx}}{2} \right)^2 + \sigma_{xy}^2 \right]^{1/2} \quad (20)$$

where

$$\begin{aligned} \frac{\sigma_{yy} - \sigma_{xx}}{2} &= -\frac{\sigma_{ox}}{2} - \sigma_y \left[B_1(1+S_1^2) \tan^{-1} \frac{2\sqrt{\frac{r_y}{r_1}} \cos \frac{\theta_1}{2}}{1 - \frac{r_y}{r_1}} + B_2(1+S_2^2) \tan^{-1} \frac{2\sqrt{\frac{r_y}{r_2}} \cos \frac{\theta_2}{2}}{1 - \frac{r_y}{r_2}} \right] \\ \sigma_{xy} &= \sigma_y \left[S_1 B_1 \log \frac{1 + \frac{r_y}{r_1} + 2\sqrt{\frac{r_y}{r_1}} \sin \frac{\theta_1}{2}}{1 + \frac{r_y}{r_1} - 2\sqrt{\frac{r_y}{r_1}} \sin \frac{\theta_1}{2}} + \frac{(1+S_2^2)^2}{4S_2} S_2 \log \frac{1 + \frac{r_y}{r_2} + 2\sqrt{\frac{r_y}{r_2}} \sin \frac{\theta_2}{2}}{1 + \frac{r_y}{r_2} - 2\sqrt{\frac{r_y}{r_2}} \sin \frac{\theta_2}{2}} \right] \end{aligned} \quad (21a)$$

and

$$\begin{aligned} r_1 &= r(\cos^2 \theta + S_1 \sin^2 \theta)^{1/2} \\ r_2 &= r(\cos^2 \theta + S_2 \sin^2 \theta)^{1/2} \\ \tan \theta_1 &= S_1 \tan \theta \\ \tan \theta_2 &= S_2 \tan \theta \end{aligned} \quad (21b)$$

If the Dugdale strip yield zone is propagating slowly, that is, at a crack velocity of $C/C_1 < 0.1$, then for all practical purposes the stress wave effect can be ignored and the static and dynamic crack tip stress fields coincide. Equations (20) and (21) can then be superimposed and, with Equations (20) through (21) can be used to construct near-field isochromatics at the tip of a propagating Dugdale strip yield zone.

The modified Dugdale model was further improved by incorporating higher order, non-singular terms in the static crack-tip stress field. This crack tip stress field contains as disposable parameters,

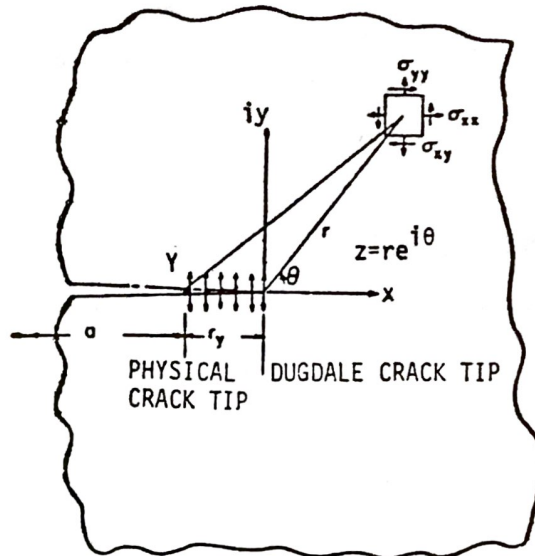


Fig.17. Dugdale strip yield zone at the crack tip of a semi-infinite crack.

the undetermined coefficients associated with the polynomial of $r^{n/2}$ and the undetermined tangential surface tractions. These disposable parameters were determined by least square fitting the isochromatics, which are associated with this modified Dugdale model, to the recorded elastic isochromatics in thin fracturing polycarbonate specimens. Figure 18 shows a set of typical experimentally observed isochromatics and those associated with the modified Dugdale model. The fractional fringe orders for the isochromatics in the yield region were estimated through the use of a constant strain-optical coefficient and deformation theory of plasticity. This and twenty other similar results verified the modified Dugdale model for rapid tearing of thin polycarbonate plates⁷⁸. The modified Dugdale model with non-singular mode II crack tip stress field was also used successfully to model the strip yield zone and the stable crack growth from a slanted crack in thin polycarbonate tension specimens⁷⁹. This modified model has been used with equal success for modeling rapid tearing of thin 7075-T6 aluminium plates.

As for a dynamic ductile fracture criterion, a critical crack tip opening displacement (CTOD) computed by the above modified Dugdale model was found to hold for both mode I and combined modes I and II rapid tearing.

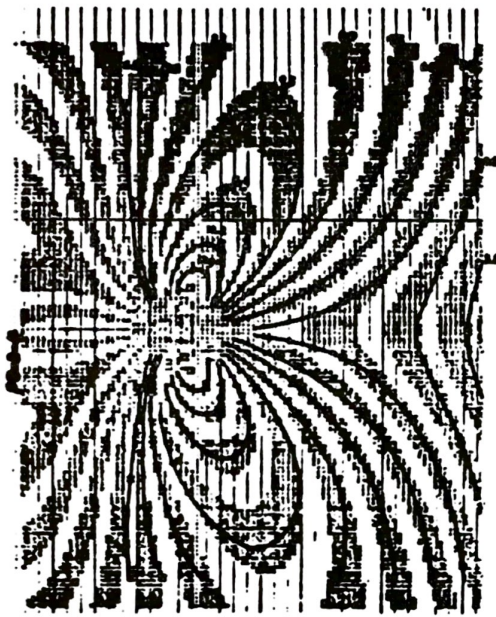


Fig.18. Experimental and theoretical isochromatics surrounding a rapidly tearing crack in a thin polycarbonate sheet

Numerical Modeling of Impact

The constitutive equations of K_{ID} versus C relation for somewhat ductile material and the critical CTOD for rapid tearing of thin ductile material can be incorporated into a dynamic finite element or finite difference codes to predict the crack propagation history in impacted structures. These numerical analyses have been restricted to the propagation of a single straight crack, such as the impacted three point bend specimen considered in Figure 12.

Another example involves the simulation of thermal shock experiment⁸⁰ of an internally preflawed A508 steel vessel, as shown in the caption of Figure 19, with an O.D. of 0.53 m and I.D. of 0.24 m. Thermal stress was induced by sudden chilling to -21°C the bore of the cylinder which was heated uniformly to 291°C . The actual crack propagation and arrest took place at a time of 150 sec and therefore the measured radial temperature distribution at this time was used to compute the pre-existing state of thermal stress using both a static finite element and finite difference code. Predetermined K_{ID} versus C relation for A508 steel was used as a constitutive equation to execute an elastodynamic finite difference code which yielded the crack propagation history of Figure 19. Figure 20 shows the stress intensity factor variations with crack extension in this cylinder. The importance

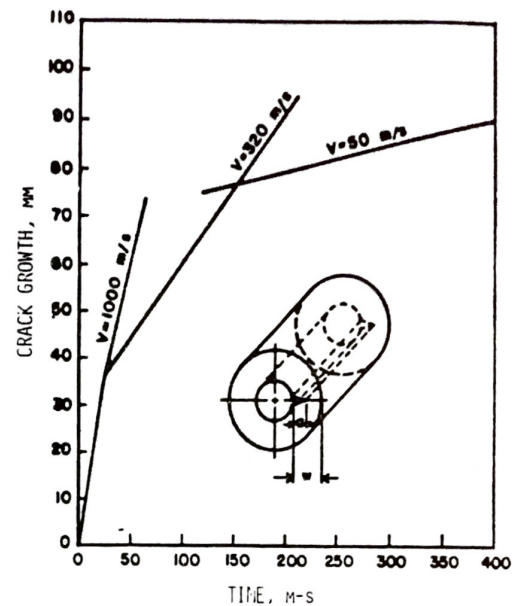


Fig.19. Crack growth vs. time for hypothetical long-jump (run arrest) event of a thermally shocked cylinder

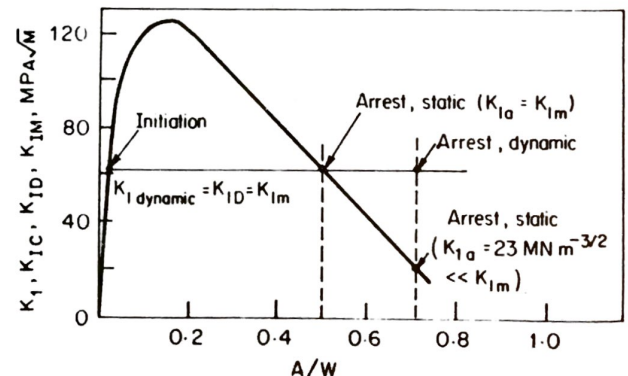


Fig.20. Analysis of hypothetical long-jump (run arrest) event of a thermally shocked cylinder

of dynamic analysis is underscored by the significantly longer arrest crack length which it predicted and demonstrated the nonconservativeness of static analysis in this study.

Another example is the crack propagation analysis of a precracked circular cylinder subjected to an impulse loading on its inner surface⁸¹. The initial flaw is located in steel cylinders with three thickness-to-mean-radius ratios of $h/R=0.017, 0.033$ and 0.085 . At time $t=0$, an impulse $I=\rho h v_0$, where ρ is the density and v is the initial radial velocity, is imposed on the bore. An elastodynamic finite difference

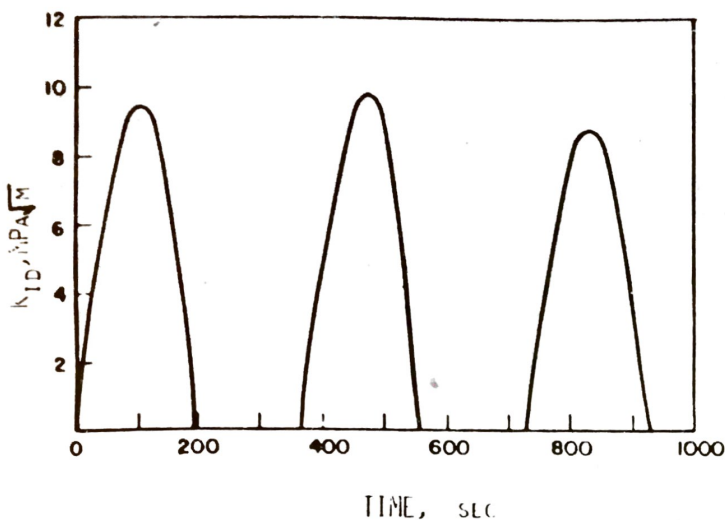


Fig.21. Stress intensity factor vs. time for a stationary crack in a steel cylinder $R=305$ mm, $h=26$ mm, $a_0=6$ mm, $v_0=1$ m/s.

code was executed with a speed independent toughness of $K_{ID}=K_{IC}$. Figure 21 shows the cyclic variation of K_I for a stationary crack subjected to an initial radial velocity of $v_0=1$ m/s at the bore. The period of the pulsating stress intensity factor is nearly equal to the breathing mode of the cylindrical oscillation of the unflawed cylinder. Figure 22 shows the crack propagation history where $K_{ID}=0.25$ MPa $\sqrt{m}$ was taken as 98 percent of the maximum stress intensity factor experienced by the stationary crack of Figure 21. The computed initial crack speed is about 300 m/s which increased to about 2200 m/s during the final stage of crack penetration.

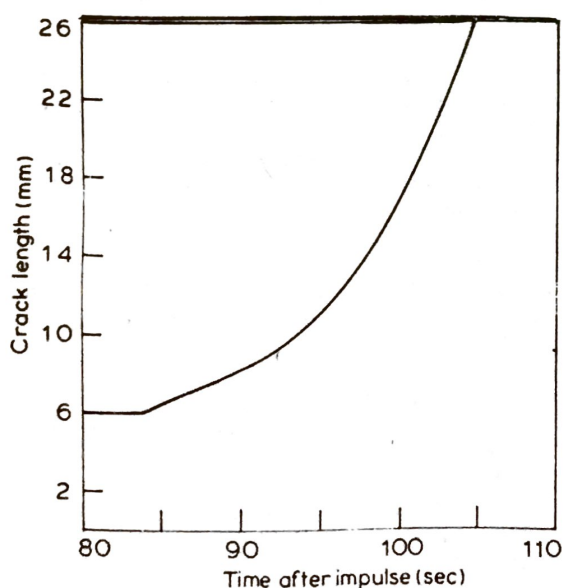


Fig.22. Crack propagation in impulsively loaded cracked steel cylinder - $R=305$ mm, $h=26$ mm, $a_0=6$ mm, $v_0=1$ m/s.

As shown above, a finite element model based on the above basic laws of dynamic fracture can be developed for predicting the dynamic fracture response of an impacted specimen. The complex multiple-crack paths in many of the impacted specimens, however, pose practical difficulties in programming, since new surface elements must be created along each crack path. A procedure for circumventing this difficulty is to replace the multiple cracks with finite elements of nil stiffnesses in directions perpendicular to the cracks. Reduction in the stiffness as well as its direction are governed by the dynamic fracture criteria discussed above and the cumbersome double nodding or node grafting procedure is eliminated. Thus, the crack is represented by a string of damaged elements. This approach in fracture mechanics, which is referred to as damage mechanics⁸², models crack opening and closure by the stress relaxation and the reduction/recovery of the stiffness of the individual elements. Frictional resistance to relative sliding of the rough surfaces of a closed crack is generated by a high friction coefficient which for soda-lime glass⁸³ varies between 0.15 and 0.5. An example of this simple modeling is shown by a finite element simulation of a structural ceramic plate impacted by a single spherical ball.

Since the stiffness of a crack body varies with the opening or closing of a crack, the material response of a crack body is nonlinear and irreversible. Moreover, the formation of a crack will result in stress relaxation and redistribution which requires an iterative algorithm to assure the convergence of the redistributed stresses which must satisfy the equilibrium condition. Damage mechanics has been used in numerical fracture analysis of brittle materials such as concrete, rock and ceramics.

The finite element model, as shown in Figure 23, of impact damage is based on an incrementally linear analysis implemented with an updated Newton-Raphson iteration scheme. This computation simulated a thick hot-pressed silicon nitride (HPSN) plate impacted by a 2.4 mm diameter tungsten carbide (WC) sphere at a velocity of 231 m/s. Table 3 shows the material properties of these two materials. Hertzian hemispherical contact pressure distribution was used as the prescribed loading.

Figure 24 shows the sectional view of the HPSN plate reported by Dao et al.⁸⁴ where qualitative agreement between numerical results generated by the finite element model and the experimental results is noted. Numerical analysis^{82,85} showed that most of the Hertzian cone cracks formed during the loading process while the lateral cracks only appeared during unloading and are in agreement with the experimental observations^{86,87}. Finite element results also indicated that the damage-induced porous zone underneath the impact site is mainly due to shear overload, which is equivalent to uniaxial compressive crushing of ceramics⁸⁸ and was induced by the high confining pressure in the contained contact region. This crushed damage zone was also observed in References 89 and 90.

Table 3
Material properties

Material	Young's Modulus E(GPa)	Poisson's Ratio	Density (kg/m ³)	Tensile Strength T(GPa)	Shear Strength S(GPa)	Friction Coefficient f
HPSN	320	0.26	3300	1.4	14	0.15
WC	700	0.24	15000	-	-	-

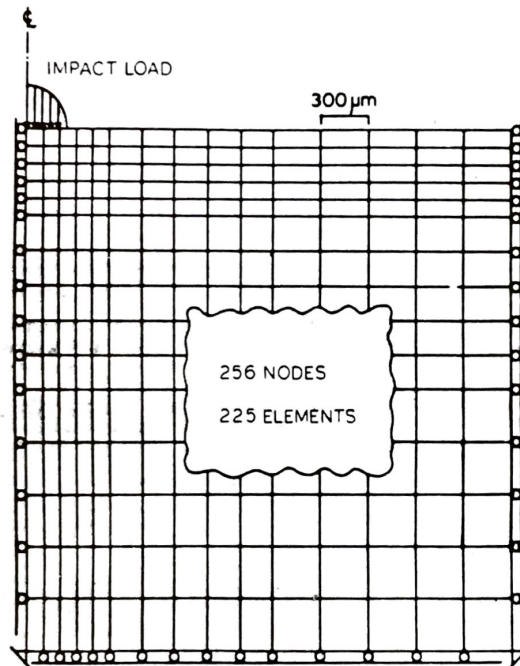


Fig.23. Finite element model of a half space subjected to normal impact loading

Conclusions

The constitutive relations for dynamic crack propagation in somewhat ductile and ductile materials were reviewed. Fracture criteria under mixed mode loading conditions for a rapidly propagating crack is presented for the non-planar crack propagation. Also, a crack branching criterion, which is useful in predicting crack bifurcation under dynamic overloading, was presented. These criteria can be incorporated into existing dynamic elasto-plastic codes and to predict the history of a single crack propagating in an impacted structure. For multiple cracking leading to fragmentation, a simple finite element code based on damage mechanics appears to be the only viable numerical tool for analysis.

Acknowledgement

The effort in compiling this paper was supported by the University of Washington. The experimental results which were generated by the authors and their colleagues, were supported by ONR, NSF, and NASA research contracts.

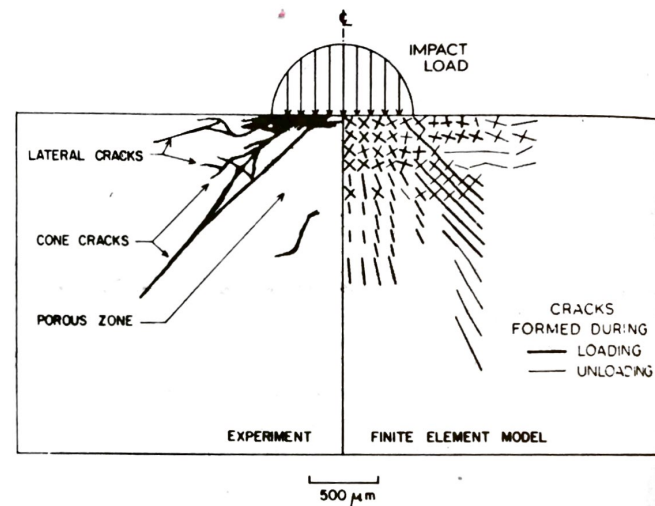


Fig.24. Damage patterns in a HPSN plate impacted by a 2.4 mm dia. WC sphere at 231 m/s

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